Deep Learning Based Sensor Fusion for Endpoint Prediction and Control of Transhumeral Prostheses

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Upper-limb prostheses can be broadly divided into three categories - cosmeses, body-powered devices, and powered devices. Cosmeses and body-powered devices are well established in clinical use and have been for over a century, but can limit functionality or lead to compensatory movements for users with limb differences. Powered devices offer the potential to replace missing degrees of freedom (DOFs), a goal that becomes more difficult the more proximal the amputation. Today most powered prostheses in clinical use are "myoelectric prostheses", and are controlled via surface electromyography (sEMG) sensors worn on the skin over muscles in the residual limb. Mostly these prostheses are limited to sequential control, meaning that the user can actively control only a single DOF at a time. This limitation arises primarily from the user's ability to voluntarily control individual muscles, which is further limited by the level of the amputation and is a particular challenge for transhumeral amputees.

Our research objective is to develop a controller for an above-elbow prosthesis, based on utilization of alternative sensor modalities for predicting hand position, specifically gaze tracking and inertial measurement units (IMUs) in conjunction with sEMG. We believe such a predictor will enable simultaneous and naturalistic control while reducing the mental load on users. To that end we have conducted healthy-subject pilot data acquisition combining sEMG, IMUs (head and arm mounted) and gaze-tracking, with optical motion capture (MoCap) for ground-truth kinematics. After basic pre-processing of the all data and conversion to a unified coordinate system, we calculate the elevation angles of both upper-arm segments according to Soecting and Flanders[1]. The first step of this work was to assess how effectively these sensor modalities can predict hand position for eventual control of a powered prosthetic arm. While IMU data was collected, for the purposes of our initial model development we assess the predictiveness of the elevation angles based on MoCap data. Future implementations will use IMU data from the upper-arm to estimate elevation angles, based on a trained model robust to variations in IMU placement.

Our current deep-learning model is a hybrid CNN and GRU (Gated Recurrent Unit), which allows us to combine the fusion capabilities of a CNN with a GRU's ability to capture temporal phenomena. This architecture was inspired by previous work where we quantified the temporal relationship between changes to a person's gaze trajectory and changes in the motion of their dominant hand. We trained the model to predict the 3D coordinates of the center of the palm along with a pair of optical markers mounted to the sides of the wrist. After training, we tested our model on unseen data, and achieved an average MAE of approximately 65mm across both hands and a number of recorded trials. We believe this accuracy is quite good given the sparsity of the training data, as it was able to estimate desired hand position to approximately 65% of the average adult male hand width (104mm wide according to [2]).

References

[1] J. F. Soechting and M. Flanders, "Errors in pointing are due to approximations in sensorimotor transformations," *Journal of Neurophysiology*, Aug. 1989. [Online]. Available: https://journals.physiology.org/doi/10.1152/jn.1989.62.2.595

[2] A. R. Tilley and H. D. Associates, *The Measure of Man and Woman: Human Factors in Design*. John Wiley & Sons, Dec. 2001, google-Books-ID: ZuoCDgAAQBAJ.